

Math 242, S16
Exam 2

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	20	
2	20	
3	24	
4	18	
5	18	
Total	100	

Good Luck !

Problem 1. (20 points) Given the following homogeneous second-order differential equation:

$$y'' + 6y' + 13y = 0.$$

(a) Find the general solution of the given differential equation.

Solution: Characteristic equation

$$r^2 + 6r + 13 = 0$$

$$\begin{aligned} r &= \frac{-6 \pm \sqrt{6^2 - 4 \cdot 1 \cdot 13}}{2} \\ &= \frac{-6 \pm \sqrt{-16}}{2} \\ &= -3 \pm 2i \end{aligned}$$

$$\text{Thus } y(x) = e^{-3x}(C_1 \cos 2x + C_2 \sin 2x)$$

(b) Find a particular solution of the equation that satisfies the initial conditions

$$y(0) = 3, \quad y'(0) = 1.$$

$$\begin{aligned} \text{Solution: } y'(x) &= (-3)e^{-3x}(C_1 \cos 2x + C_2 \sin 2x) \\ &\quad + e^{-3x}(-2C_1 \sin 2x + 2C_2 \cos 2x) \\ &= e^{-3x}[C_1(-3 \cos 2x - 2 \sin 2x) + \\ &\quad C_2(-3 \sin 2x + 2 \cos 2x)] \end{aligned}$$

$$\begin{aligned} y(0) = 3 &\Rightarrow C_1 = 3 \\ y'(0) = 1 &\Rightarrow -3C_1 + 2C_2 = 1 \Rightarrow \begin{cases} C_1 = 3 \\ C_2 = 5 \end{cases} \end{aligned}$$

$$\text{Thus } y(x) = e^{-3x}(3 \cos 2x + 5 \sin 2x)$$

Problem 2. (20 points) Given a nonhomogeneous linear third-order differential equation:

$$y^{(3)} + 4y'' + 4y' = 6x - 1.$$

(a) Find the complementary solution y_c of the above nonhomogeneous equation (i.e., the general solution of the associated homogeneous equation).

Solution: Characteristic equation of the homogeneous equation.

$$r^3 + 4r^2 + 4r = 0$$

$$r(r^2 + 4r + 4) = 0$$

$$r(r+2)^2 = 0$$

$$r_1 = 0, r_2 = r_3 = -2$$

$$\begin{aligned} \Rightarrow y_c(x) &= C_1 + (C_2 + C_3 x)e^{-2x} \\ &= C_1 + C_2 e^{-2x} + C_3 x e^{-2x} \end{aligned}$$

(b) Then find a particular solution y_p of the nonhomogeneous equation by the method of undetermined coefficient.

Solution: We have $f(x) = 3x - 1$, thus

$$y_p(x) = x^s (Ax + B)$$

$\rightarrow s=1$ to avoid duplication of terms in $y_p(x)$

$$\Rightarrow y_p(x) = x(Ax + B) = Ax^2 + Bx$$

$$y_p'(x) = 2Ax + B$$

$$y_p''(x) = 2A$$

$$y_p'''(x) = 0$$

$$\begin{aligned} \Rightarrow y_p^{(3)} + 4y_p'' + 4y_p' &= 0 + 4(2A) + 4(2Ax + B) \\ &= 8Ax + (8A + 4B) \equiv 6x - 1 \end{aligned}$$

$$\Rightarrow \begin{cases} 8A = 6 \\ 8A + 4B = -1 \end{cases} \Rightarrow \begin{aligned} A &= \frac{3}{4} \\ B &= -\frac{7}{4} \end{aligned}$$

$$\text{thus } y_p(x) = \frac{3}{4}x^2 - \frac{7}{4}x$$

Problem 3. (24 points) Find the Laplace or inverse Laplace transforms of the functions:

(a) $\mathcal{L}\{t^3 - 3 \sinh 2t\}$

$$= \frac{3!}{s^4} - 3 \cdot \frac{2}{s^2 + 4} = \frac{6}{s^4} - \frac{6}{s^2 + 4}$$

(b) $\mathcal{L}\{t \cos 4t\}$ using the formula $\mathcal{L}\{-tf(t)\} = F'(s)$ where $F(s) = \mathcal{L}\{f(t)\}$.

$$f(t) = \cos 4t$$

$$\mathcal{L}\{t \cos 4t\} = -\frac{d}{ds} [\mathcal{L}\{\cos 4t\}]$$

$$= -\frac{d}{ds} \left[\frac{s}{s^2 + 16} \right]$$

$$= -\frac{1 \cdot (s^2 + 16) - s \cdot (2s)}{(s^2 + 16)^2}$$

$$= -\frac{16 - s^2}{(s^2 + 16)^2}$$

(c) $\mathcal{L}^{-1}\left\{\frac{2}{s^2(s^2 + 4)}\right\}$ using the formula $\mathcal{L}^{-1}\{F(s)G(s)\} = f(t) * g(t)$ where $f(t) * g(t) =$

$$\int_0^t f(\tau)g(t-\tau) d\tau, \mathcal{L}^{-1}\{F(s)\} = f(t) \text{ and } \mathcal{L}^{-1}\{G(s)\} = g(t).$$

$$F(s) = \frac{1}{s^2} \Rightarrow f(t) = t, \quad G(s) = \frac{2}{s^2 + 4} \Rightarrow g(t) = \sin 2t$$

Thus

$$\mathcal{L}^{-1}\left\{\frac{2}{s^2(s^2 + 4)}\right\} = f(t) * g(t) = \int_0^t \tau \sin 2(t-\tau) d\tau$$

$$= \left[\frac{1}{2} \tau \cos 2(t-\tau) \right]_{\tau=0}^{\tau=t} - \int_0^t \frac{1}{2} \cos 2(t-\tau) d\tau$$

$$= \frac{1}{2} t - \left[-\frac{1}{4} \sin 2(t-\tau) \right]_{\tau=0}^{\tau=t}$$

$$= \frac{1}{2} t - \left[0 + \frac{1}{4} \sin 2\tau \right]$$

$$= \frac{1}{2} t - \frac{1}{4} \sin 2\tau$$

Problem 4. (18 points) Use Laplace transform to solve the initial value problem:

$$x'' + 9x = 0; \quad x(0) = 3, \quad x'(0) = 6.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\begin{aligned}\mathcal{L}\{x''(t)\} &= s^2 X(s) - sx(0) - x'(0) \\ &= s^2 X(s) - 3s - 6\end{aligned}$$

Then

$$\mathcal{L}\{x'' + 9x\} = \mathcal{L}\{0\}$$

$$\mathcal{L}\{x''\} + 9\mathcal{L}\{x\} = 0$$

$$s^2 X(s) - 3s - 6 - 9X(s) = 0$$

$$X(s) = \frac{3s+6}{s^2+9}$$

$$\Rightarrow x(t) = \mathcal{L}^{-1}\left\{\frac{3s+6}{s^2+9}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{3s}{s^2+9}\right\} + \mathcal{L}^{-1}\left\{\frac{6}{s^2+9}\right\}$$

$$= 3\cos 3t + 2\sin 3t$$

Problem 5. (18 points) Use Laplace transform to solve the initial value problem

$$x'' + 2x' - 8x = 2; \quad x(0) = x'(0) = 0.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\mathcal{L}\{x'(t)\} = sX(s) - x(0) = sX(s)$$

$$\mathcal{L}\{x''(t)\} = s^2X(s) - sx(0) - x'(0) = s^2X(s)$$

Then

$$\mathcal{L}\{x'' + 2x' - 8x\} = \mathcal{L}\{2\}$$

$$\mathcal{L}\{x''\} + 2\mathcal{L}\{x'\} - 8\mathcal{L}\{x\} = \frac{2}{s}$$

$$s^2X(s) + 2sX(s) - 8X(s) = \frac{2}{s}$$

$$X(s) = \frac{2}{s(s^2 + 2s - 8)}$$

$$\begin{aligned} \frac{2}{s(s^2 + 2s - 8)} &= \frac{2}{s(s-2)(s+4)} = \frac{A}{s} + \frac{B}{s-2} + \frac{C}{s+4} \\ &= \frac{A(s-2)(s+4) + Bs(s+4) + Cs(s-2)}{s(s-2)(s+4)} \end{aligned}$$

$$\text{let } s=0, \text{ we get } 2 = A(-2)(4) \Rightarrow A = -\frac{1}{4}$$

$$s=2, \text{ we get } 2 = B(2)(6) \Rightarrow B = \frac{1}{6}$$

$$s=-4, \text{ we get } 2 = C(-4)(-6) \Rightarrow C = \frac{1}{12}$$

Thus

$$\begin{aligned} x(t) &= \mathcal{L}^{-1}\left\{\frac{2}{s(s^2 + 2s - 8)}\right\} \\ &= \mathcal{L}^{-1}\left\{-\frac{1}{4} \frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{6} \frac{1}{s-2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{12} \frac{1}{s+4}\right\} \\ &= -\frac{1}{4} + \frac{1}{6}e^{2t} + \frac{1}{12}e^{-4t} \end{aligned}$$